

Quantitative Research Methods

Explainable AI with Shapley Values

Advanced module A2 — optional self-study

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Where this module fits

This is one of four **optional advanced modules** that extend the core course. Nothing here is exam material for the main chapters — but everything here is built from tools you already own.

You already know	From	Used here for
Flexibility vs. interpretability	Ch. 2	why accurate models need explaining
Boosting, variable importance	Ch. 8	the black box we explain; the benchmark
Deep networks	Ch. 10	the archetypal black box (same machinery)
The 2^p subset count	Ch. 0b, Ch. 6	the cost of exact Shapley computation
Seeded simulation, resampling	Ch. 5	background samples, Monte-Carlo error

Today's question

Chapter 8 gave us a boosted model that predicts well; Chapter 10 gave us networks nobody can read. This module answers: **for one prediction, how much did each feature contribute — exactly, with a guarantee?**

Contents

- 1 Why Explainability
- 2 The Shapley Value
- 3 Games to Models
- 4 Computation
- 5 Reading the Output
- 6 Pitfalls
- 7 Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this module

Symbol	Meaning
p	number of features (game-theory: players); here $p = 6$
$\mathcal{P} = \{1, \dots, p\}$	the grand coalition: all players together
$S \subseteq \mathcal{P} \setminus \{j\}$	a coalition (subset of players) not containing j ; $ S $ its size
$v(S)$	value function: the worth achieved by coalition S alone
φ_j	Shapley value of player / feature j
$f(x)$	the fitted model's prediction at input x
x_S	the explained instance's values on the features in S
$X_{\bar{S}}$	the random features <i>outside</i> S
φ_0	baseline: $E[f(X)]$, the average prediction
$b^{(1)}, \dots, b^{(B)}$	background sample of B rows (here $B = 100$)
$\pi, \text{pre}_\pi(j)$	a random ordering of the players; those placed before j in π
m	number of sampled orderings in Monte-Carlo estimation

Sources and references

Primary textbook (course background)

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python*. Springer. — Chapters 8 and 10 supply the models we explain.

Module sources

- Shapley, L. S. (1953). A value for n -person games. *Contributions to the Theory of Games II*, Princeton UP.
- Štrumbelj, E., & Kononenko, I. (2014). Explaining prediction models ... with feature contributions. *Knowledge and Information Systems*.
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NeurIPS* (the SHAP paper).
- Lundberg, S. M., et al. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence* (TreeSHAP).
- Aas, K., Jullum, M., & Løland, A. (2021). Explaining individual predictions when features are dependent. *Artificial Intelligence*.
- Molnar, C. (2022). *Interpretable Machine Learning*, 2nd ed. (free online book).

Learning objectives

By the end of this module you will be able to:

- **Distinguish** local from global explanations, and say what a feature attribution should promise.
- **Define** the Shapley value of a cooperative game and **verify** its four axioms on concrete numbers.
- **Translate** a fitted model into a game via the value function $v(S)$, and **defend** a choice between marginal and conditional expectations.
- **Implement** exact Shapley values from scratch (2^P enumeration) and their Monte-Carlo approximation, and **check** the efficiency axiom numerically.
- **Read** waterfall, global-importance and dependence displays, and **compare** them with Chapter 8's permutation importance.
- **Avoid** the three classic pitfalls: off-manifold evaluation, causal over-reading, and attribution instability.

Roadmap of this module

From a parlour game to a regulator-ready explanation

- 1 Why accurate models need explaining; local vs. global.
- 2 Cooperative games and the Shapley value: formula, axioms, a fully worked three-player game.
- 3 From games to models: the value function $v(S)$ and the marginal-vs.-conditional choice.
- 4 Computation: exact 2^p enumeration by hand, Monte-Carlo sampling, KernelSHAP and TreeSHAP.
- 5 Reading the output: waterfall, mean $|\varphi_j|$, dependence plots — against permutation importance.
- 6 Pitfalls: correlated features, causation, instability — and what industry regulation demands.

Where this module is used in industry

Sector	Concrete application	Which decision it drives
Consumer credit	Adverse-action reasons for each declined loan application	Which “principal reasons” appear on the legally required notice
Insurance	Justifying a boosted pricing model factor by factor	Whether a rating factor survives actuarial and regulatory review
Healthcare	Which vitals and labs drove a patient’s risk score	Whether the clinician trusts, overrides or escalates the alert
Fraud & AML	Explaining why a transaction was flagged	Investigator triage; evidence in the case file
Tech / marketing	Debugging ranking and churn models: what drove a bad score	Fix the data pipeline vs. retrain the model
Model risk / audit	Global driver analysis and drift monitoring	Model approval, re-validation, de-commissioning

In industry — The common thread

In every row the model is a *tree ensemble or neural net* that beats the interpretable alternative — and a human downstream who must *account* for a single prediction. Shapley values are the de-facto standard answer because they come with a guarantee: the attributions **sum exactly** to the prediction.

Industry case in depth: explaining a credit decline

In industry — The setting: a boosted model decides — the law wants reasons

A bank scores consumer-loan applications with a gradient-boosted model — hundreds of trees over ~ 40 features. US regulation (ECOA / Regulation B) requires the bank to give each declined applicant the **principal reasons**; the EU AI Act classifies credit scoring as high-risk with transparency duties. “The model said so” is not a legal answer.

How Shapley values enter the pipeline

For each declined applicant x the bank computes attributions $\varphi_1, \dots, \varphi_p$ that split $f(x) - E[f(X)]$ across features. The most negative φ_j 's are mapped to standardised reason codes (“debt-to-income ratio too high”). The *background* is chosen deliberately — e.g. approved applicants — so the reasons answer “why you, compared to those who were approved”.

Two honest caveats — both taught in this module

The reasons describe the *model*, not the applicant's causal reality — they are not financial advice. And with correlated features (income, debt, limits) the split between related reasons is partly a modelling choice, so banks fix background, seed and model version in their model-risk documentation.

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Accurate models nobody can read

Chapters 8 and 10 ended with a trade: **accuracy up, readability gone**.

Our running model (built in this module)

A GradientBoostingRegressor predicting **log salary** on the Hitters data ($n = 263$ players after dropping missing salaries, $p = 6$ features) reaches **training** $R^2 = 0.965$ — using 100 trees of depth 3. That is roughly 800 leaf rules added together: no coefficient table to read, no single split to point at.

What we can no longer answer (yet)

- A player asks: *why does the model pay me less than him?*
- A manager asks: *which statistics drive salary overall?*
- An auditor asks: *does the model use the feature you claim it ignores?*

Takeaway

We refuse to trade the accuracy back. Instead we add an **explanation layer** on top of the fixed black box f — this module builds that layer with a guarantee.

Local vs. global — and what an attribution must promise

Local explanation

For **one** instance x : how much did each feature value push *this* prediction $f(x)$ away from the average prediction? (*the declined applicant, the flagged transaction*)

Global explanation

For the **whole** model: which features matter most *on average* across all instances? (*model validation, driver analysis*)

What a feature attribution should mean

Numbers $\varphi_1, \dots, \varphi_p$, one per feature, such that: (i) they **sum exactly** to $f(x) - E[f(X)]$ — no leftover, no double count; (ii) a feature the model **ignores** gets 0; (iii) two features used **identically** get the same number. These wishes will become the Shapley *axioms* — and they pin down a *unique* answer.

Takeaway

Shapley values are **local** objects; averaging $|\varphi_j|$ over many instances then buys the global picture. One machinery, both questions.

Exercise A2.1 — Local or global? [Concept]

Exercise

For each question, say whether it needs a **local** or a **global** explanation, and what the attribution numbers should sum to (if anything):

- ① “Why was *my* loan declined?”
- ② “Which features should we monitor for drift in production?”
- ③ “The model predicts salary \$743k for this veteran while the average prediction is \$349k — what accounts for the gap?”
- ④ “Can we drop PutOuts from the pipeline without losing accuracy?”

Solution — Exercise A2.1

Solution

Method. Ask *whose* prediction is being discussed: one instance $\Rightarrow$ local; the model as a whole $\Rightarrow$ global.

- 1 **Local.** One applicant, one prediction. The attributions should sum to $f(x) - E[f(X)]$: the gap between *this* score and the average score.
- 2 **Global.** A property of the model over its input distribution; mean $|\varphi_j|$ across instances is a natural monitor. No sum constraint — it is an average of magnitudes.
- 3 **Local**, and the sum is pinned down exactly: on the log scale the φ_j must add up to $\log 743 - \log 349 \approx 0.756$. (This is our running example — we will compute those six numbers exactly.)
- 4 **Global** — but note the trap: a small mean $|\varphi_j|$ says the model *currently* leans little on PutOuts, not that the feature is useless; after refitting without it, other correlated features may absorb its role.

Common mistake

Answering a local question with a global tool: telling the declined applicant “income is the most important feature of our model” explains the *model*, not *their* decision.

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Cooperative games: players, coalitions, value

A **cooperative game** is a pair $(\mathcal{P}, v)$:

Definition

- $\mathcal{P} = \{1, \dots, p\}$: a set of p **players**;
- $v : 2^{\mathcal{P}} \rightarrow \mathbb{R}$ with $v(\emptyset) = 0$: a **value function** giving the worth $v(S)$ that any **coalition** $S \subseteq \mathcal{P}$ can achieve on its own.

The question: how should the grand coalition's worth $v(\mathcal{P})$ be **divided fairly** among the players?

How to read this

“Fair” must reflect *contribution*: a player who adds a lot to every coalition they join deserves more. The natural raw material is the **marginal contribution** $v(S \cup \{j\}) - v(S)$ — what j adds when joining the coalition S . The difficulty: this number *depends on* S .

In industry — Cost allocation: the original use

Shapley's 1953 idea long predates machine learning: utilities split shared infrastructure costs, airports split runway costs across airline types, and firms allocate IT platform costs to business units — all with the same formula we are about to build.

A three-player game with concrete numbers

Three specialists found a consultancy: **A** (modeller), **B** (engineer), **C** (designer). Quarterly revenue (in 1 000 €) for every possible team:

The full value function

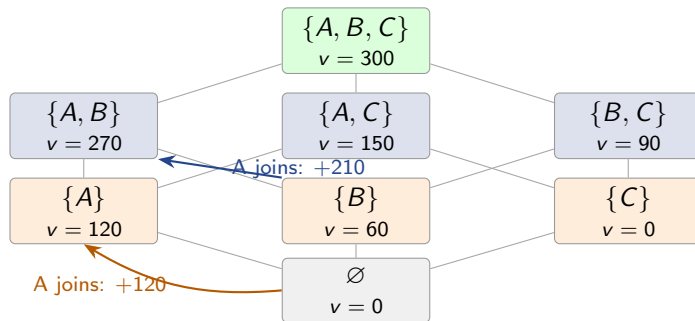
S	$\emptyset$	$\{A\}$	$\{B\}$	$\{C\}$	$\{A, B\}$	$\{A, C\}$	$\{B, C\}$	$\{A, B, C\}$
$v(S)$	0	120	60	0	270	150	90	300

C earns *nothing* alone, yet every team gains 30 when C joins it ($150 - 120$, $90 - 60$, $300 - 270$): C's design work only pays off on top of a product. A and B together show *synergy*: $270 > 120 + 60$.

Takeaway

Naive splits fail: equal thirds (100 each) ignores that A alone is worth 120; paying stand-alone values (120, 60, 0) only distributes 180 of the 300 and gives C nothing despite C's reliable +30. We need a rule that **averages marginal contributions over all ways a team can form**.

The coalition lattice (native diagram)



Takeaway

Every edge is a marginal contribution. Player A adds +120 when joining the empty room but +210 when joining B (270 – 60): contributions **depend on who is already there**. The Shapley value will average over *all* such edges, correctly weighted.

Exercise A2.2 — Fair division in the consultancy game [Math]

Exercise

Use the consultancy game ($v = 0, 120, 60, 0, 270, 150, 90, 300$ in the table order above). A player's Shapley value averages their marginal contribution over the six possible **arrival orders** $ABC, ACB, BAC, BCA, CAB, CBA$ (each equally likely).

- 1 Compute player A's marginal contribution in each of the six orders, and average them to get φ_A .
- 2 Repeat for B and C (or use any shortcut you can justify).
- 3 Verify **efficiency**: $\varphi_A + \varphi_B + \varphi_C = v(\{A, B, C\})$.
- 4 Is C a *dummy* player? Justify with one coalition.

Solution — Exercise A2.2 (1/2)

Solution — the six arrival orders

Method. In an arrival order, each player's contribution is the jump in v when they walk in. Tabulate all six orders once; then every player's Shapley value is a column average of the same table.

Order	A adds	B adds	C adds
$A B C$	120	$270 - 120 = 150$	$300 - 270 = 30$
$A C B$	120	$300 - 150 = 150$	$150 - 120 = 30$
$B A C$	$270 - 60 = 210$	60	$300 - 270 = 30$
$B C A$	$300 - 90 = 210$	60	$90 - 60 = 30$
$C A B$	$150 - 0 = 150$	$300 - 150 = 150$	0
$C B A$	$300 - 90 = 210$	$90 - 0 = 90$	0
mean	$\varphi_A = \frac{1020}{6} = 170$	$\varphi_B = \frac{660}{6} = 110$	$\varphi_C = \frac{120}{6} = 20$

Solution — Exercise A2.2 (2/2)

Solution — efficiency and the dummy check

Part 3 — efficiency.

$$\varphi_A + \varphi_B + \varphi_C = 170 + 110 + 20 = 300 = v(\{A, B, C\}). \checkmark$$

The whole revenue is distributed — nothing left over, nothing invented.

Part 4 — is C a dummy? No. A dummy adds nothing to *any* coalition. But $v(\{A, C\}) - v(\{A\}) = 150 - 120 = 30 \neq 0$: C adds 30 to every non-empty team. C's stand-alone value is 0, yet C's Shapley pay is $20 > 0$ — the value correctly rewards *team* contribution, not solo performance.

Comparison. Equal thirds would overpay C (100 vs. 20); stand-alone pay would underpay C (0) and fail efficiency ($180 \neq 300$). The order-averaged split is the only one we will be able to *axiomatise*.

Common mistake

Computing marginal contributions only for the order A, B, C and calling it a day. A single order is arbitrary — A would get 120 instead of 170 — and the result would depend on the labelling of the players.

The Shapley value: the formula

Rule (Shapley, 1953)

The Shapley value of player j in the game $(\mathcal{P}, v)$ is

$$\varphi_j = \sum_{S \subseteq \mathcal{P} \setminus \{j\}} \frac{|S|! (p - |S| - 1)!}{p!} (v(S \cup \{j\}) - v(S))$$

Symbol by symbol

- $S \subseteq \mathcal{P} \setminus \{j\}$ — every coalition j could join (all 2^{p-1} subsets not containing j); $v(S \cup \{j\}) - v(S)$ — j 's **marginal contribution** on arrival.
- $|S|!$ — arrival orders of S *before* j ; $(p - |S| - 1)!$ — orders of the rest *after* j ; $p!$ — all orders. So the weight is exactly $\Pr(j \text{ arrives right after the set } S)$ under a uniformly random order: the formula *is* Exercise A2.2's order-averaging, written without listing orders.
- Equivalent form: weight $= \frac{1}{p} \binom{p-1}{|S|}^{-1}$ — pick j 's arrival position first (each size equally likely), then the predecessors uniformly among that size.

Checking the formula on the consultancy game

$\varphi_A=170$ — now via the subset formula ($p=3$)

Weights: $|S| = 0 : \frac{0!2!}{3!} = \frac{1}{3}; \quad |S| = 1 : \frac{1!1!}{3!} = \frac{1}{6}; \quad |S| = 2 : \frac{2!0!}{3!} = \frac{1}{3}.$

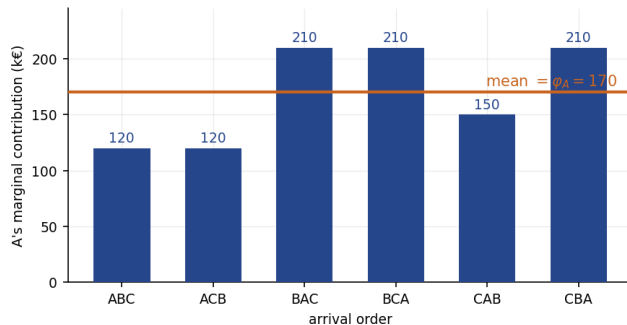
S	weight	$v(S \cup \{A\})$	$v(S)$	term
$\emptyset$	1/3	120	0	40
$\{B\}$	1/6	270	60	35
$\{C\}$	1/6	150	0	25
$\{B, C\}$	1/3	300	90	70
			φ_A	170

Same 170 as the six-order average — with 4 terms instead of 6 orders.

Takeaway

Endpoint coalitions carry the **largest** weight ($\frac{1}{3}$ each): one way to be first or last, $\binom{p-1}{|S|}$ ways to see a middle-sized crowd. For $p = 6$: $\frac{1}{6}, \frac{1}{30}, \frac{1}{60}, \frac{1}{60}, \frac{1}{30}, \frac{1}{6}$ by size — and for each j the weights **sum to 1**: φ_j is a weighted *average* of marginal contributions.

The Shapley value is an average, drawn



Takeaway

A's marginal contribution in each of the $3! = 6$ arrival orders of the consultancy game. The Shapley value is nothing more mysterious than the *mean of these bars* — which is also why sampling orders (Monte-Carlo, later this module) estimates it unbiasedly.

The four axioms — and the uniqueness theorem

Each axiom in one honest sentence

- 1 **Efficiency:** the shares add up exactly, $\sum_{j=1}^P \varphi_j = v(\mathcal{P}) - v(\emptyset)$ — the whole surplus is distributed.
- 2 **Symmetry:** if $v(S \cup \{i\}) = v(S \cup \{j\})$ for every S containing neither, then $\varphi_i = \varphi_j$ — equal treatment of interchangeable players.
- 3 **Dummy (null player):** if $v(S \cup \{j\}) = v(S)$ for every S , then $\varphi_j = 0$ — who never adds anything gets nothing.
- 4 **Additivity:** for two games on the same players, $\varphi_j(v + w) = \varphi_j(v) + \varphi_j(w)$ — playing two games at once changes nothing about either.

Why this matters

Theorem (Shapley, 1953). The Shapley value is the *unique* allocation satisfying all four axioms. So once we accept the four wishes from the motivation slide, there is **no freedom left** — the formula is not one heuristic among many, it is the only answer.

Exercise A2.3 — Weights and the dummy axiom [Math]

Exercise

Work directly with the weight $w(S) = \frac{|S|! (p - |S| - 1)!}{p!}$.

- ❶ For $p = 3$, list $w(S)$ for all four coalitions $S \subseteq \{1, 2, 3\} \setminus \{j\}$ and show they sum to 1.
- ❷ Prove the **dummy axiom** from the formula: if $v(S \cup \{j\}) = v(S)$ for all S , then $\varphi_j = 0$.
- ❸ Using $w(S) = \frac{1}{p} \binom{p-1}{|S|}^{-1}$, explain in one sentence why the empty and the almost-full coalition carry the largest weight.

Solution — Exercise A2.3

Solution

Part 1 — the four weights for $p = 3$. The subsets of the other two players are $\emptyset$, two singletons, and the pair:

$$w = \frac{0!2!}{3!}, \frac{1!1!}{3!}, \frac{1!1!}{3!}, \frac{2!0!}{3!} = \frac{1}{3} + \frac{1}{6} + \frac{1}{6} + \frac{1}{3} = 1. \checkmark$$

Part 2 — dummy. Every term of $\varphi_j = \sum_S w(S) (v(S \cup \{j\}) - v(S))$ contains the factor $v(S \cup \{j\}) - v(S)$, which is 0 for every S by assumption; a sum of zeros is 0. (Because the weights sum to 1, the same argument shows more: φ_j always lies between j 's smallest and largest marginal contribution.)

Part 3 — endpoint weights. Writing $w(S) = \frac{1}{p} \binom{p-1}{|S|}^{-1}$: each size $|S|$ is equally likely ($1/p$), and that probability is shared equally among the $\binom{p-1}{|S|}$ coalitions of that size — sizes 0 and $p-1$ have only *one* coalition each, so each keeps the full $1/p$.

Common mistake

Reading the weight as “all 2^{p-1} coalitions are equally important”. They are not: for $p = 6$ a single middle coalition carries weight $\frac{1}{60}$, the empty one $\frac{1}{6}$ — ten times more.

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Features are players: the value function of a prediction

Fix the trained model f and one instance x to explain. Let the **features be the players**; a coalition S means “these feature values are *known* to be x_S ”.

Rule (SHAP value function, marginal form)

$$v_x(S) = E[f(x_S, X_{\bar{S}})] - E[f(X)] \approx \frac{1}{B} \sum_{b=1}^B f(x_S, x_S^{(b)}) - \varphi_0$$

where $\varphi_0 = E[f(X)]$ and the expectation is estimated on a **background sample** $x^{(1)}, \dots, x^{(B)}$.

How to read this

- $f(x_S, X_{\bar{S}})$: evaluate the model with the **known** features pinned at x 's values and the **unknown** ones drawn from data — an average prediction given what we know.
- $v_x(\emptyset) = 0$ and $v_x(\mathcal{P}) = f(x) - \varphi_0$: knowing nothing gives the average prediction; knowing everything gives $f(x)$.
- **Efficiency** then reads $\sum_j \varphi_j = f(x) - \varphi_0$: the attributions decompose *this* prediction's distance from the average — Lundberg & Lee (2017) call this *local accuracy*, and SHAP = *SHapley Additive exPlanations*.

Marginal or conditional? The one modelling choice that matters

Two ways to “not know” the features outside S :

Marginal (interventional)

$$v_x(S) = E[f(x_S, X_{\bar{S}})] - \varphi_0$$

Draw $X_{\bar{S}}$ from its own distribution, **ignoring** its correlation with x_S — like an intervention on the model’s inputs.

Conditional (observational)

$$v_x(S) = E[f(X) \mid X_S = x_S] - \varphi_0$$

Draw $X_{\bar{S}}$ from its distribution **given** $X_S = x_S$ — what we would expect from the data.

Each option’s honest price

Marginal respects the *model*: unused features are true dummies ($\varphi_j = 0$), but the average runs over feature combinations the data never produces (off-manifold — pitfalls section). **Conditional** respects the *data*: no impossible points, but correlated proxies of a used feature get credit even if the model never touches them — the dummy axiom fails *by design*, and the conditional distributions must themselves be estimated.

Takeaway

This module (like most software defaults) uses the **marginal** form: exact dummies, easy estimation — and we will look its off-manifold problem in the eye.

Our testbed: gradient boosting on Hitters

- **Data:** Hitters — 322 baseball players (1986/87 season); 59 missing salaries dropped $\Rightarrow n = 263$.
- **Target:** $\log(\text{Salary})$ (salaries in \$1 000, range 67.5–2 460; the log tames the right tail).
- **Features** ($p = 6$): Years, CHits, CRBI, Walks, Hits, PutOuts — small enough for exact $2^6 = 64$ -coalition Shapley.
- **Model:** GradientBoostingRegressor (sklearn defaults, `random_state=2024`): training $R^2 = 0.965$.
- **Background:** $B = 100$ rows drawn with `default_rng(2024)`; baseline $\varphi_0 = 5.854$ log-dollars $\approx \$349\text{k}$.

The instance we will explain

Row 188: a **24-season veteran** holding the career-hits record — Years = 24, CHits = 4 256, CRBI = 1 314, Walks = 30, Hits = 52, PutOuts = 523. Actual 1987 salary: \$750k. Model: $f(x) = 6.610 \approx \$743\text{k}$.

Run this live in the lab notebook

Section §1–§2 of `advanced_02_shapley_lab.ipynb` loads Hitters, fits this exact model and builds the background sample — run it now so the later cells have f , bg and $base$ in memory.

Exercise A2.4 — The duplicated feature [Concept]

Exercise

A bank's dataset accidentally contains `income` twice (`income_a = income_b` for every customer). The fitted model happens to split **only** on `income_a`.

- 1 Under the **marginal** value function, what is $\varphi_{\text{income}_b}$? Which axiom decides it?
- 2 Under the **conditional** value function, why does `income_b` receive a non-zero share?
- 3 Which of the two constructions evaluates the model at customer profiles that cannot exist (`income_a` $\neq$ `income_b`)?
- 4 Give one question for which the marginal answer is the right one, and one for which the conditional answer is.

Solution — Exercise A2.4

Solution

Part 1. $\varphi_{\text{income}_b} = 0$ by the **dummy axiom**: the model never touches `income_b`, so pinning it changes no prediction, so every marginal contribution is 0.

Part 2. Conditioning on `income_b = x` forces `income_a = x` too (they are identical), so $E[f(X) \mid \text{income}_b]$ moves exactly as much as conditioning on `income_a` — informationally the twins are symmetric, and they share the credit. The dummy axiom fails, *deliberately*.

Part 3. The **marginal** one: computing $v(\{\text{income}_a\})$ it pins `income_a` at the customer's value while drawing `income_b` from the background — creating profiles with two different incomes, which the data manifold does not contain.

Part 4. “Does the deployed model discriminate on `income_b`?” — marginal (mechanistic: no, it cannot). “Does the score reveal / depend on the information in `income_b`?” — conditional (informational: yes, fully).

Common mistake

Believing one of the two is “the correct Shapley value”. They are exact answers to *different questions* — the choice of v is part of the explanation and must be reported with it.

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Python lab — run it live in the notebook

Companion notebook: `advanced_02_shapley_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook (it sits beside this deck) and run it cell by cell:

- §1, *Data and model* — Hitters, log salary, the boosted model;
- §2, *The value function* — background sample, $v_x(S)$, the baseline;
- §3, *Exact Shapley by enumeration* — all 64 coalitions, efficiency check;
- §4, *Monte-Carlo permutation sampling* — convergence to the exact values;
- §5, *Local and global pictures* — waterfall, mean $|\varphi_j|$, dependence;
- §6, *Pitfalls* — correlated features and retrain instability.

Data loads via the ISLP package or the bundled CSV files; *Lecture exercises — worked Python solutions* and open exercises end the notebook.

Exact enumeration costs 2^p — count before you compute

The formula for one φ_j touches all 2^{p-1} coalitions; all p features together need every one of the 2^p values $v_x(S)$, each an average over B background rows.

The same count as best-subset selection (Ch. 0b and Ch. 6)

	$p = 6$ (here)	$p = 20$	$p = 50$
coalitions 2^p	64	1 048 576	$\approx 1.1 \times 10^{15}$
model calls at $B = 100$	6 400	$\approx 10^8$	hopeless

Chapter 0b's counting slide and Chapter 6's best-subset explosion are *literally the same* 2^p : “every feature in or out”. There we gave up on exhaustive search; here we can afford it only because $p = 6$.

Takeaway

Exact Shapley is a **small- p luxury**: 6 400 predictions per instance at $p = 6$ (about 1.7 million for all 263 players — seconds on a laptop). Beyond $p \approx 15$ –20 ($2^{15} = 32\,768$ coalitions) we must sample — next slides.

Exact Shapley by hand (1): model and value function

```
import itertools, numpy as np, pandas as pd # std stack
from math import factorial
from sklearn.ensemble import GradientBoostingRegressor

df = pd.read_csv("Hitters.csv").dropna().reset_index(drop=True)
feats = ["Years", "CHits", "CRBI", "Walks", "Hits", "PutOuts"]
X = df[feats].to_numpy(float) # (263, 6)
y = np.log(df["Salary"]).to_numpy() # log $1000s
f = GradientBoostingRegressor(random_state=2024).fit(X, y)
print(round(f.score(X, y), 3)) # train R^2: 0.965

rng = np.random.default_rng(2024) # house seed
bg = X[rng.choice(len(X), 100, replace=False)] # background, B=100
base = f.predict(bg).mean() # phi_0 = E[f]: 5.854

def v(S, x): # marginal value function
    Z = bg.copy() # B background rows
    if S: Z[:, list(S)] = x[list(S)] # pin known features to x
    return f.predict(Z).mean() - base # average prediction - phi_0
```

Exact Shapley by hand (2): the sum over all coalitions

```
p = len(feats) # p = 6
w = {s: factorial(s) * factorial(p - s - 1) / factorial(p)
      for s in range(p)} # weight by |S|

def exact_shapley(x): # x: one row, shape (6,)
    phi = np.zeros(p)
    for j in range(p): # one player at a time
        rest = [k for k in range(p) if k != j]
        for r in range(p): # coalition sizes 0..p-1
            for S in itertools.combinations(rest, r):
                phi[j] += w[r] * (v(S + (j,), x) - v(S, x))
    return phi

x_star = X[188] # the 24-season veteran
phi = exact_shapley(x_star)
print(dict(zip(feats, np.round(phi, 4)))) # one number per feature
fx = f.predict(x_star.reshape(1, -1))[0] # f(x*) = 6.6102
print(round(phi.sum(), 4), round(fx - base, 4)) # efficiency check
```

Run this live in the lab notebook

Notebook §3 runs exactly this code — `itertools.combinations` over all 64 coalitions — and times it, so you see the 2^P cost with your own clock.

The efficiency axiom, verified to machine precision

Output of the previous slide

	Years	CHits	CRBI	Walks	Hits	PutOuts
φ_j	-0.5245	+0.5626	+0.5474	-0.0452	-0.1084	+0.3244

Sum of attributions: $\sum_j \varphi_j = 0.7564$. Prediction minus baseline: $f(x^*) - \varphi_0 = 6.6102 - 5.8538 = 0.7564$.

Takeaway

Efficiency holds **exactly**, not approximately: across all 263 players the largest violation of $\sum_j \varphi_j = f(x) - \varphi_0$ is 4.4×10^{-15} — floating-point noise. This is the property no rival attribution method (gradients, single-feature ablations) gives you for free.

In industry — Model risk: the audit test

Validation teams re-run exactly this check on sampled instances: if reported attributions do not reproduce $f(x) - \varphi_0$ to numerical precision, the explanation pipeline (wrong background, stale model version) is flagged before the numbers reach a customer letter.

Monte-Carlo: sample arrival orders instead of enumerating

Algorithm (permutation sampling; Štrumbelj & Kononenko, 2014)

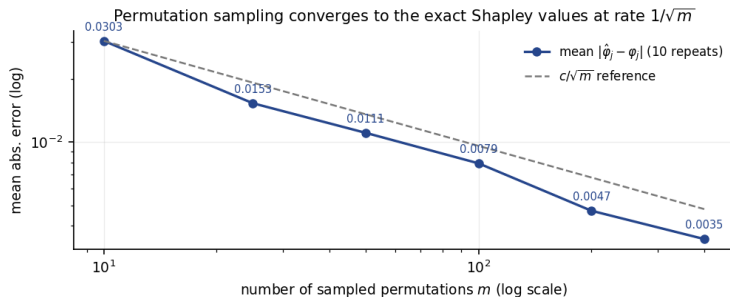
Repeat for $t = 1, \dots, m$: draw a uniformly random ordering π_t of the p features; walk through it, pinning one feature at a time, and award each feature the change in v_x it causes on arrival. Average:

$$\hat{\varphi}_j = \frac{1}{m} \sum_{t=1}^m \left(v_x(\text{pre}_{\pi_t}(j) \cup \{j\}) - v_x(\text{pre}_{\pi_t}(j)) \right).$$

How to read this

- **Unbiased:** the exact φ_j is the expectation of the arrival-order contribution — sampling orders estimates it.
- **Error** $\propto 1/\sqrt{m}$ (a plain Monte-Carlo mean): quadruple the orders, halve the error.
- **Cost:** $m \cdot p$ evaluations of v_x , i.e. $m p B$ model calls — independent of 2^p . At $p = 6$, $m = 100$: 60 000 calls, — *more* than exact (6 400)! Sampling only pays once 2^p explodes.
- Efficiency now holds only *in expectation*; each run leaves a small random residual.

Convergence: sampled orders vs. the exact answer



Takeaway

For the veteran instance, $m = 25$ sampled orders already bring the mean absolute error to 0.015 log-dollars ($\sim 2\%$ of the largest $|\varphi_j|$); by $m = 400$ it is 0.0035. The dashed $c/\sqrt{m}$ line confirms the rate: $\times 4$ orders $\Rightarrow$ error halved (0.0153 $\rightarrow$ 0.0079 from $m = 25$ to 100).

Scaling up in practice: KernelSHAP, TreeSHAP, shap

KernelSHAP in one idea (Lundberg & Lee, 2017)

Efficiency suggests a regression: fit $v_x(S) \approx \sum_{j \in S} \varphi_j$ by **weighted least squares** over sampled coalitions S , with the *Shapley kernel* weight $\pi(S) = \frac{p-1}{\binom{p}{|S|} |S| (p-|S|)}$. With *all* 2^p coalitions this WLS reproduces the exact Shapley values; with a sample it approximates them — efficiently, because each sampled coalition informs all p features at once.

TreeSHAP: exactness for free on tree ensembles

For trees, the expectations $E[f(x_S, X_{\bar{S}})]$ can be pushed *through the tree structure*: exact Shapley values in polynomial time (roughly $\# \text{trees} \times \# \text{leaves} \times \text{depth}^2$ per instance) — no 2^p , no sampling noise. This is why boosted trees + SHAP became the industry pairing.

In industry — The production tool

The `shap` package ships all of this (`TreeExplainer`, `KernelExplainer`, the plots). We built everything from scratch so that `shap`'s output is a convenience, not a mystery — it is not installed on the course machines, but one `pip install shap` away in Colab.

Extended Exercise A2.1 — Monte-Carlo Shapley from scratch [Python]

Extended exercise (15 min)

Implement permutation sampling for the veteran instance x^* (row 188) and quantify its accuracy against the exact values from the previous slides.

- 1 Write `mc_shapley(x, m, rng)`: sample m orderings with `rng.permutation(p)`; for each, walk through the features, awarding each its change in v (start the walk at $v(\emptyset) = 0$).
- 2 Run it with $m = 100$ and `default_rng(2024)`; tabulate $\hat{\varphi}_j$ against the exact φ_j and report the mean and maximum absolute error.
- 3 Estimate the error at $m \in \{25, 100, 400\}$ (average over 10 repeats). Does quadrupling m halve the error?
- 4 Compare total model calls of your $m = 100$ run against exact enumeration at $p = 6$. Which wins here, and when does that flip?

Run this live

The worked solution sits in `advanced_02_shapley_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise A2.1 — Monte-Carlo Shapley from scratch [Python]*.

Solution — Extended Exercise A2.1 (1/3)

Solution — code: sampling arrival orders

```
def mc_shapley(x, m, rng): # estimate all p values
    est = np.zeros(p)
    for _ in range(m): # m random arrival orders
        perm = rng.permutation(p) # e.g. [3 0 5 1 4 2]
        S, v_prev = [], 0.0 # empty room: v = 0
        for j in perm: # features walk in
            v_new = v(tuple(S) + (int(j),), x) # value after j joins
            est[j] += v_new - v_prev # j's marginal contribution
            v_prev = v_new # room now includes j
            S.append(int(j))
    return est / m # average over orders

rng = np.random.default_rng(2024) # fresh seeded generator
est = mc_shapley(x_star, 100, rng) # m = 100 orders
```

Method. One pass through an ordering costs p evaluations of v and awards every feature exactly once — so each pass is one telescoping sample of *all* p Shapley values simultaneously.

Solution — Extended Exercise A2.1 (2/3)

Solution — the $m=100$ run against the exact values

	Years	CHits	CRBI	Walks	Hits	PutOuts
$\hat{\varphi}_j$ ($m=100$)	-0.5164	+0.5408	+0.5741	-0.0496	-0.1071	+0.3145
φ_j (exact)	-0.5245	+0.5626	+0.5474	-0.0452	-0.1084	+0.3244

Part 2 — errors. Mean absolute error = 0.012, maximum = 0.027 (on CRBI) — about 2–5% of the leading attributions: every sign and the full ranking are already correct at $m = 100$.

Part 3 — rate. Averaged over 10 repeats: error 0.0153 at $m = 25$, 0.0079 at $m = 100$, 0.0035 at $m = 400$. Each quadrupling roughly halves the error — the $1/\sqrt{m}$ signature of a Monte-Carlo mean.

Solution — Extended Exercise A2.1 (3/3)

Solution — cost accounting and when sampling wins

Part 4 — model calls.

- **Exact** at $p = 6$: $2^p \cdot B = 64 \times 100 = 6\,400$ calls.
- **MC** at $m = 100$: $m \cdot p \cdot B = 100 \times 6 \times 100 = 60\,000$ calls — *nine times more*, for an approximate answer.

At $p = 6$, exact enumeration wins outright. The flip is fast: exact doubles with every added feature ($2^{20} \cdot B \approx 10^8$ calls at $p = 20$), while the MC budget $m p B$ grows only *linearly* in p — at $p = 20$ the same $m = 100$ costs 200 000 calls, now 500× cheaper than exact.

Take-away. Enumerate while you can ($p \lesssim 15$); sample when you must; and if the model is a tree ensemble, TreeSHAP gives exactness at neither price.

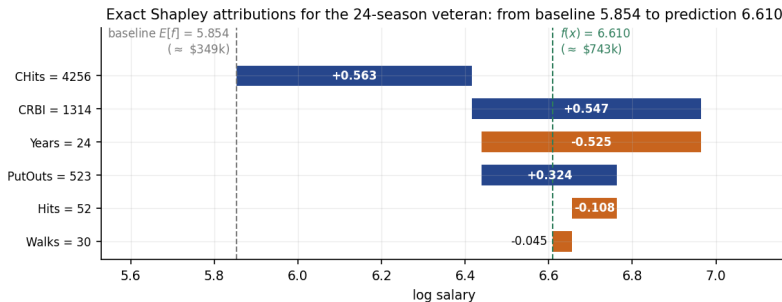
Common mistake

Judging convergence from a single run at one m : a lucky draw can look converged while another seed disagrees in the second decimal. Always repeat with several seeds (or track the standard error across orderings) before trusting $\hat{\varphi}_j$.

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Local view: the waterfall of one prediction



Takeaway

Read it left to right: start at the baseline 5.854 ($\approx \$349k$), add the six attributions in order of size, land exactly on $f(x) = 6.610$ ($\approx \$743k$). The record career totals push up (+0.563, +0.547); being 24 seasons old pushes down (-0.525); a weak current season (52 hits) subtracts another -0.108.

Reading attributions on the log scale

Our model predicts *log* salary, so attributions are additive in logs — i.e. **multiplicative** in dollars.

The veteran's salary — rebuilt factor by factor

$$\underbrace{349}_{\text{baseline}} \times \underbrace{e^{+0.563}}_{\text{CHits: } \times 1.76} \times \underbrace{e^{+0.547}}_{\text{CRBI: } \times 1.73} \times \underbrace{e^{-0.525}}_{\text{Years: } \times 0.59} \times \underbrace{e^{+0.324}}_{\text{PutOuts}} \times \underbrace{e^{-0.108-0.045}}_{\text{Hits, Walks}} \approx 743 \text{ (\$1 000)}$$

All six together: $e^{0.756} = 2.13$ — this player is priced at **2.1 times** the baseline salary, and the waterfall says exactly why.

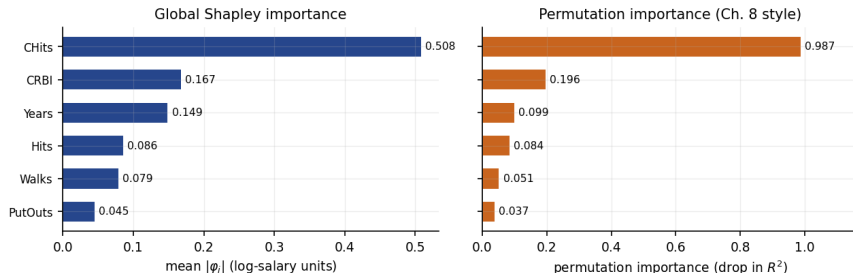
Units checklist before you present

(i) *Which output?* log salary, probability, log-odds, €? (ii) *Which baseline?* $\varphi_0 = 5.854$ is the mean prediction over the $B = 100$ background rows — a different background moves every φ_j . (iii) *Sanity sum:* $\varphi_0 + \sum_j \varphi_j$ must equal $f(x)$.

Run this live in the lab notebook

Notebook §5 rebuilds this waterfall and lets you swap instance and background — watch the baseline and every bar move together.

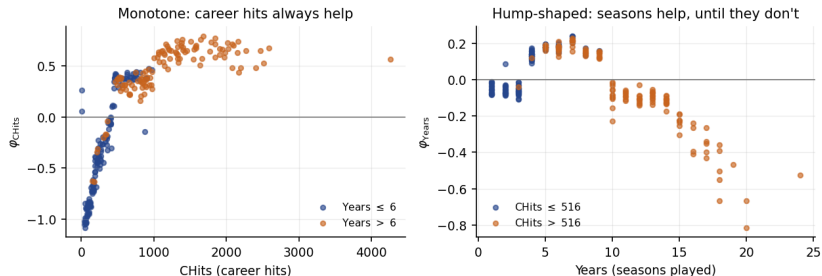
Global view: mean $|\varphi_j|$ vs. permutation importance



Takeaway

Both rankings agree here: CHits dominates, then CRBI and Years. But the *numbers* answer different questions: mean $|\varphi_j|$ (0.508 for CHits) is an average **contribution in log-salary units**; permutation importance (0.987) is the **drop in R^2** when the feature is scrambled (Ch. 8). Under correlation they can diverge: permuting CHits also destroys information shared with CRBI ($r = 0.947$), inflating its apparent solo importance.

Dependence view: how an attribution varies with its feature



Takeaway

Left: ϕ_{CHits} rises monotonically — mean -0.70 below 250 career hits, $+0.63$ above 1000: career volume always pays. Right: ϕ_{Years} is **hump-shaped** — $+0.19$ around seasons 5–7, falling to -0.45 beyond 15: *given* career totals, extra seasons mean less production per season. Vertical spread at a fixed x -value (and the colour split) is the fingerprint of **interactions**.

Exercise A2.5 — Explain another player [Python]

Exercise

Row 0 of the cleaned data is a 14-season veteran: `Years = 14`, `CHits = 835`, `CRBI = 414`, `Walks = 39`, `Hits = 81`, `PutOuts = 632`; actual salary \$475k. With the fitted model and `exact_shapley` from the lab:

- 1 Compute $f(x)$ and the six exact attributions φ_j .
- 2 Verify efficiency: $\sum_j \varphi_j = f(x) - \varphi_0$.
- 3 Name the largest positive and the largest negative contribution and restate each as a multiplicative salary factor e^{φ_j} .

Run this live

The worked solution sits in `advanced_02_shapley_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Exercise A2.5 — Explain another player [Python]*.

Solution — Exercise A2.5

Solution

Part 1 — the numbers. $f(x) = 6.167$ log-dollars $\approx$ \$477k (actual: \$475k — the model is nearly on the money here).

	CHits	PutOuts	CRBI	Walks	Hits	Years
φ_j	+0.287	+0.176	+0.112	−0.037	−0.100	−0.125

Part 2 — efficiency. $\sum_j \varphi_j = 0.313$ and $f(x) - \varphi_0 = 6.167 - 5.854 = 0.313$: equal to machine precision. ✓

Part 3 — reading. Largest positive: CHits = +0.287, a factor $e^{0.287} \approx 1.33$ — a solid career record raises the predicted salary by a third. Largest negative: Years = −0.125, factor $e^{-0.125} \approx 0.88$ — the same late-career discount we saw in the dependence plot (14 seasons sits on its downslope), only milder than the record-holder's −0.525 at 24 seasons.

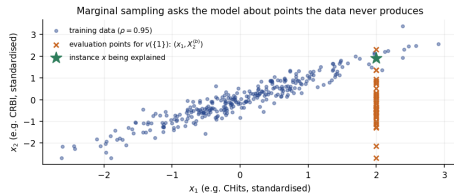
Common mistake

Comparing φ_j across explanations computed with *different backgrounds*: attributions are shares of $f(x) - \varphi_0$, so changing φ_0 rescales the whole decomposition. One report, one background.

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Pitfall 1: correlated features put the model off-manifold



Takeaway

Marginal sampling pins x_1 and draws x_2 from the background — the orange crosses. With $\rho = 0.95$ most sit **outside** the data ridge: we query the model where it never saw data (in Hitters, $r_{\text{CHits}, \text{CRBI}} = 0.947$, so $v(S)$ averages over impossible careers like 4 256 hits with 3 RBI). The attributions inherit whatever the model *extrapolates* out there.

Common mistake

Trusting fine differences between correlated features' attributions: that split rests exactly on the off-manifold region. Robust practice: report the **group attribution** $\varphi_{\text{CHits}} + \varphi_{\text{CRBI}}$, or switch to a conditional value function and accept its price (Exercise A2.4).

Pitfall 2: an explanation is not a cause

$\varphi_{\text{Years}} = -0.525$ for the veteran. Three readings — only one is licensed:

What the number does and does not say

- ✓ **Model-mechanical:** pinning $\text{Years} = 24$ (vs. background values) lowers the model's average output by 0.525 — a statement about f and the background, nothing else.
- × **Worldly-causal:** “playing fewer seasons would have raised his salary” — Shapley intervenes on the *model's inputs*, not on the world; no design (randomisation, instruments) supports this.
- × **Recourse:** “to be paid more, reduce Years ” — feature values cannot be moved independently in reality (Years drags CHits , $r = 0.898$), and the model would be refitted on any new regime anyway.

Takeaway

Shapley explanations answer “**how did the model use its inputs**”, never “what should the customer change”. Recourse and causal claims need causal tools — a different module, different assumptions.

Pitfall 3: attributions move when the model is retrained

Same data and accuracy — different explanations

Two stochastic refits (subsample=0.7, seeds 1 and 2) both reach train $R^2 \approx 0.96$. Exact Shapley for the veteran:

	φ_{CHits}	φ_{CRBI}	sum of the pair
refit, seed 1	+0.702	+0.521	+1.223
refit, seed 2	+0.600	+0.545	+1.146

φ_{CHits} swings by 0.10 ($\sim 15\%$) between two equally accurate models; the **pair's sum** moves far less. Correlated features are interchangeable to the fit — so the fit chooses arbitrarily, and the attribution follows.

In industry — Model risk & regulation: pinning the explanation down

Under SR 11-7-style model governance and the EU AI Act, banks version *everything the explanation depends on*: model artefact, background sample, seed, and shap version. A declined customer who reapplies must not receive contradictory “principal reasons” because a nightly retrain shuffled credit between two collinear features.

Extended Exercise A2.2 — Designing a credit-explanation pipeline

[Integrative]

Extended exercise (15 min)

A bank declines consumer loans with a gradient-boosted classifier ($p = 40$ features; `income`, `debt_to_income`, `n_open_lines`, `history_length` form a correlated cluster). Regulation requires the top-3 “principal reasons” on every decline notice. Design the pipeline:

- 1 Local or global explanations for the notice — and what do you keep the global ones for?
- 2 Choose and defend a **background**: all applicants, approved applicants, or a single “average customer” row.
- 3 Exact, permutation-sampling, KernelSHAP or TreeSHAP — pick one and justify with the cost arithmetic of this module.
- 4 The correlated cluster: one concrete danger for the reason codes, and one mitigation.
- 5 The applicant replies: “so if I raise my income by €5 000 I will be approved?” — draft the honest two-sentence answer.

Solution — Extended Exercise A2.2 (1/2)

Solution — parts 1–3: the architecture

Part 1 — local for the notice, global for governance. The notice explains *one* decision $\Rightarrow$ local φ_j for that applicant; the top-3 negative attributions map to standardised reason codes. Global mean $|\varphi_j|$ goes to validation: driver stability over time, drift alarms, fair-lending review.

Part 2 — background = approved applicants. The legal question is “why were *you* declined *rather than approved*”, so the reference population should be the approved pool: φ_0 becomes the average score of approved customers and negative φ_j ’s read “what separates you from them”. (All-applicants blurs the question; a single average row is not a distribution at all and makes the explanation hostage to one synthetic point.)

Part 3 — TreeSHAP. $p = 40$ kills exact enumeration ($2^{40} \approx 10^{12}$ coalitions); permutation sampling and KernelSHAP cost $m p B$ model calls *per applicant* with sampling noise on top — awkward when notices must be reproducible. The model is a tree ensemble, so TreeSHAP is exact, fast and deterministic: the only defensible choice here.

Solution — Extended Exercise A2.2 (2/2)

Solution — parts 4–5: correlation and the customer's question

Part 4 — the cluster danger. Credit for the debt burden can split arbitrarily across `income`, `debt_to_income`, `n_open_lines`: after a routine retrain the top-3 list can swap “income too low” for “too many open lines” with *no change in the applicant* — inconsistent notices, regulatory exposure. **Mitigation:** define reason codes on **feature groups** (sum the cluster's φ_j 's — efficiency makes group sums well-defined), and freeze model + background + seed per reporting period.

Part 5 — the honest answer. “The reasons describe how our current model weighed your data compared with approved customers; they are not a promise that changing one number changes the decision. A higher income would enter a new application, but the outcome depends on your whole profile at that time.”

Common mistake

Publishing reason codes from raw per-feature attributions in a correlated feature space. The maths is exact, but the *split within a cluster* is a modelling artefact — group first, then explain.

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Module A2 in one slide

What we accomplished

Exact, axiom-backed feature attributions for a black-box model — built entirely from scratch.

- Local vs. global explanations, and what an attribution must promise.
- Cooperative games: value functions, marginal contributions, a fully worked three-player division (170/110/20).
- The Shapley value: the $\frac{|S|!(p-|S|-1)!}{p!}$ formula, four axioms, uniqueness.
- Models as games: $v_x(S)$ via background samples; marginal vs. conditional.
- Computation: exact 2^p enumeration ($p = 6$: 64 coalitions), $1/\sqrt{m}$ Monte-Carlo, KernelSHAP, TreeSHAP.
- Reading output: waterfall (349k $\rightarrow$ 743k), mean $|\varphi_j|$ vs. permutation importance, dependence plots.
- Pitfalls: off-manifold sampling, explanation $\neq$ causation, retrain instability.

Key formulas at a glance

Shapley value $\varphi_j = \sum_{S \subseteq \mathcal{P} \setminus \{j\}} \frac{|S|! (p - |S| - 1)!}{p!} (v(S \cup \{j\}) - v(S)).$

Weight identity $\frac{|S|! (p - |S| - 1)!}{p!} = \frac{1}{p} \binom{p-1}{|S|}^{-1} = \Pr(j \text{ arrives right after } S); \text{ weights sum to } 1.$

Order form $\varphi_j = E_\pi [v(\text{pre}_\pi(j) \cup \{j\}) - v(\text{pre}_\pi(j))] — \text{basis of Monte-Carlo sampling, error } \propto 1/\sqrt{m}.$

Model as game $v_x(S) = E[f(x_S, X_{\bar{S}})] - \varphi_0$ (marginal) vs. $E[f(X) | X_S = x_S] - \varphi_0$ (conditional).

Local accuracy $f(x) = \varphi_0 + \sum_{j=1}^p \varphi_j$ (efficiency, applied to the prediction game).

Global importance $I_j = \frac{1}{n} \sum_{i=1}^n |\varphi_j^{(i)}|$ (mean absolute attribution).

KernelSHAP kernel $\pi(S) = \frac{p-1}{\binom{p}{|S|} |S| (p-|S|)}$ (WLS over coalitions recovers φ).

Vocabulary checklist

Term	Meaning
Coalition S	subset of players / features treated as “present”
Value function $v(S)$	worth of a coalition; for models: expected prediction given x_S
Marginal contribution	$v(S \cup \{j\}) - v(S)$: what j adds on arrival
Shapley value φ_j	weighted average of j ’s marginal contributions
Efficiency / local accuracy	attributions sum exactly to $f(x) - \varphi_0$
Dummy axiom	unused player gets 0
Baseline φ_0	average prediction over the background sample
Background sample	data rows standing in for “unknown” features
Marginal vs. conditional	break vs. respect feature correlations in $v(S)$
Permutation sampling	Monte-Carlo Shapley from random arrival orders
KernelSHAP / TreeSHAP	WLS approximation / exact polynomial-time trees
Waterfall plot	one prediction rebuilt attribution by attribution
Mean $ \varphi_j $	global importance from local attributions
Off-manifold evaluation	querying the model at impossible feature combinations

Decision rules of thumb

- $p \lesssim 15$ and a cheap model $\Rightarrow$ **enumerate exactly**; verify efficiency to machine precision as a pipeline test.
- Tree ensemble $\Rightarrow$ **TreeSHAP** (exact, deterministic); anything else at large $p \Rightarrow$ KernelSHAP or permutation sampling with repeats and reported Monte-Carlo error.
- Choose the **background to match the question** (“compared to whom?”) — and keep it fixed across every explanation you compare.
- Correlated feature clusters $\Rightarrow$ report **group sums** of attributions; distrust the within-cluster split.
- Always state the **output scale** (log, probability, log-odds) and the baseline next to any φ_j .
- Say “the *model* used x_j this way” — never “ x_j *caused* the outcome” and never “change x_j to change the decision”.

Common pitfalls

Don't

- ❶ Read Shapley values causally or hand them to customers as recourse advice.
- ❷ Compare attributions computed against different backgrounds, model versions or seeds.
- ❸ Trust the credit split *within* a correlated cluster (off-manifold arithmetic decides it).
- ❹ Report sampled Shapley values without a Monte-Carlo error estimate.
- ❺ Confuse mean $|\varphi_j|$ with permutation importance — different units, different question.
- ❻ Treat the explanation as a property of reality — it is a property of *one fitted model* plus *one value-function choice*.

Self-check questions

Can you answer these without the slides?

- 1 Why is the Shapley weight $|S|!(p - |S| - 1)!/p!$ exactly the probability that j arrives right after the coalition S ?
- 2 State the four axioms; which one turns Shapley values into an exact decomposition of $f(x) - \varphi_0$?
- 3 In the consultancy game, why does C earn 20 despite $v(\{C\}) = 0$?
- 4 What breaks if you estimate $v_x(S)$ with marginal sampling when $r_{\text{CHits}, \text{CRBI}} = 0.947$?
- 5 You quadruple the number of sampled orderings. What happens to the estimation error, and why?
- 6 Why did φ_{CHits} change from 0.702 to 0.600 across two retrains — and what stayed (almost) put?

Takeaway

If question 4 or 6 feels shaky, re-read the Pitfalls section — that is where explanation projects fail in practice.

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the module: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later project sends you back here.

Contents of the appendix

- **The weights, derived** — the counting argument behind $|S|!(p - |S| - 1)!/p!$; moved out because the main deck only needs the *reading* of the weight, not its proof.
- **Efficiency, proved** — the telescoping argument; two lines, but a proof nonetheless.
- **KernelSHAP as weighted regression** — the algebra sketch behind the one-slide summary in the main deck.
- **Estimating conditional value functions** — why the “respect-the-data” option is genuinely hard; for readers heading into research or high-stakes deployment.
- **Drill exercise (glove game)** — one more full Shapley computation for exam practice, with solution.

The weights, derived from random arrival orders

Claim

Under a uniformly random ordering π of $\mathcal{P}$,

$$\Pr(\text{pre}_\pi(j) = S) = \frac{|S|! (p - |S| - 1)!}{p!} \quad \text{for any } S \subseteq \mathcal{P} \setminus \{j\}.$$

The counting

An ordering with $\text{pre}_\pi(j) = S$ is built by: arranging the $|S|$ members of S before j ($|S|!$ ways), then j (fixed), then the remaining $p - |S| - 1$ players ($(p - |S| - 1)!$ ways). Total favourable orderings: $|S|! (p - |S| - 1)!$ out of $p!$ equally likely ones. ■

Takeaway

Hence $\varphi_j = \sum_S \Pr(\text{pre}_\pi(j) = S) (v(S \cup \{j\}) - v(S)) = E_\pi[j\text{'s arrival contribution}]$: the subset formula and the order-average are the *same object*, which is why permutation sampling is unbiased.

Efficiency, proved in two lines

Claim

$$\sum_{j=1}^p \varphi_j = v(\mathcal{P}) - v(\emptyset).$$

Telescoping

Fix any ordering $\pi = (j_1, \dots, j_p)$ and sum the arrival contributions *along this one ordering*:

$$\sum_{k=1}^p [v(\{j_1, \dots, j_k\}) - v(\{j_1, \dots, j_{k-1}\})] = v(\mathcal{P}) - v(\emptyset),$$

because every intermediate $v(\cdot)$ appears once with $+$ and once with $-$. The sum is the same for *every* ordering, and averaging over π (which is what $\sum_j \varphi_j$ does, by the previous slide) preserves it. ■

Takeaway

Symmetry and dummy are equally quick from the order view; only *uniqueness* needs real work — see Shapley (1953).

KernelSHAP as a weighted regression (sketch)

Represent a coalition S by $z \in \{0, 1\}^p$ and posit the **additive surrogate** $g(z) = \varphi_0 + \sum_j \varphi_j z_j$.

The estimator

$$\hat{\varphi} = \arg \min_{\varphi} \sum_z \pi(z) (v_x(z) - g(z))^2, \quad \pi(z) = \frac{p-1}{\binom{p}{|z|} |z| (p-|z|)},$$

with $g(0) = 0$ and $g(1) = f(x) - \varphi_0$ imposed exactly (the kernel is $+\infty$ there, so they act as constraints).

How to read this

- Lundberg & Lee (2017) show: with *all* 2^p coalitions, the WLS solution equals the Shapley values — the kernel is reverse-engineered so that this holds.
- In practice one samples coalitions $\propto \pi(z)$; each sampled $v_x(z)$ informs *all* p coefficients simultaneously — the efficiency advantage over order sampling.
- Caveat: sampled KernelSHAP is still Monte-Carlo — report its error, as with permutation sampling.

Estimating conditional value functions is genuinely hard

The conditional choice $v_x(S) = E[f(X) \mid X_S = x_S] - \varphi_0$ needs the conditional law of $X_{\bar{S}}$ given $X_S = x_S$ — for every S .

The estimation menu (Aas et al. 2021)

- **Empirical / kernel:** reweight background rows by closeness to x_S — struggles beyond a few dimensions.
- **Parametric:** assume Gaussian (or copula) features, condition in closed form — clean, but a real assumption about the data.
- **Generative / surrogate:** train a model of $X_{\bar{S}} \mid X_S$ — now the *explanation* has its own model risk.
- **Path-dependent TreeSHAP:** approximates conditioning with the tree's split cover statistics — fast, but neither cleanly marginal nor cleanly conditional.

Takeaway

This is why the marginal form is the pragmatic default: its estimator is a plain average. The price is Pitfall 1 — choose which price you pay, and **say so in the report**.

Exercise A2.6 — The glove game (drill) [Math]

Exercise

Players 1 and 2 each own a **left** glove; player 3 owns a **right** glove. A coalition earns 1 if it holds at least one complete pair (a left *and* a right glove), else 0: $v(S) = 1$ iff $3 \in S$ and $S \cap \{1, 2\} \neq \emptyset$.

- 1 Write out $v(S)$ for all eight coalitions.
- 2 Compute φ_3 with the subset formula ($p = 3$ weights $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$).
- 3 Deduce φ_1 and φ_2 *without* further computation, naming the axioms you use.

Solution — Exercise A2.6

Solution

Part 1. $v = 0$ for $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}$; $v = 1$ for $\{1, 3\}, \{2, 3\}, \{1, 2, 3\}$.

Part 2 — φ_3 via the four coalitions $S \subseteq \{1, 2\}$:

$$\varphi_3 = \underbrace{\frac{1}{3}(0-0)}_{S=\emptyset} + \underbrace{\frac{1}{6}(1-0)}_{S=\{1\}} + \underbrace{\frac{1}{6}(1-0)}_{S=\{2\}} + \underbrace{\frac{1}{3}(1-0)}_{S=\{1,2\}} = \frac{1}{6} + \frac{1}{6} + \frac{1}{3} = \frac{2}{3}.$$

Part 3. Players 1 and 2 are interchangeable in v ($v(S \cup \{1\}) = v(S \cup \{2\})$ for every S avoiding both), so **symmetry** gives $\varphi_1 = \varphi_2$; **efficiency** gives $\varphi_1 + \varphi_2 + \varphi_3 = v(\mathcal{P}) = 1$, hence $\varphi_1 = \varphi_2 = \frac{1}{2}(1 - \frac{2}{3}) = \frac{1}{6}$.

Take-away. Scarcity is worth money: the single right glove earns $\frac{2}{3}$, four times each plentiful left glove — the value prices *replaceability*, exactly what it will later do with correlated features.

Common mistake

Splitting 1 as $\frac{1}{3}$ each “because there are three players”: equal split ignores that removing player 1 barely matters ($\{2, 3\}$ still earns 1) while removing player 3 zeroes the game.